VOICE PITCHER / 98 x 98 mm / TOP ASSEMBLY

STM32H743 + PCM3060 / stereo line I/O / USB-C device / SSD1306

0.1 engineering prototype. Targets are not measured specifications.

MCU

File: mcu.kicad_sch

POWER

File: power.kicad_sch

USB

File: usb.kicad_sch

CODEC

File: codec.kicad_sch

ANALOG

File: analog.kicad_sch

UI

File: ui.kicad_sch

ENGINEERING PROTOTYPE — NOT VALIDATED ON HARDWARE

Xenoah

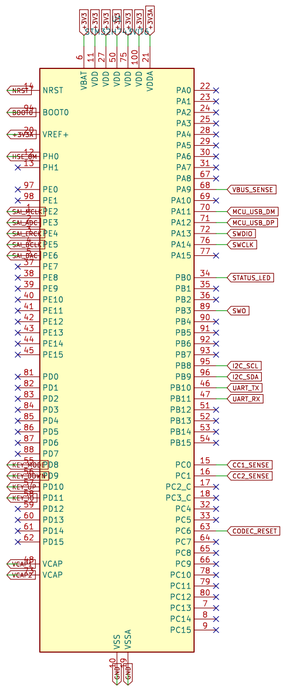
Sheet: /
File: voice_pitcher.kicad_sch

Title: VOICE PITCHER — ROOT

Size: A4	Date: 2026-11-01	**Rev: 0.1-prototype**
KiCad E.D.A. 9.0.2		Id: 1/7

All named nets are global. Unused pins are explicitly NC.

See docs/hardware.md for design assumptions and open validation items.







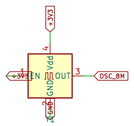




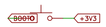
















ENGINEERING PROTOTYPE — NOT VALIDATED ON HARDWARE

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Sheet: /MCU/

File: mcu.kicad_sch

Title: VOICE PITCHER — MCU

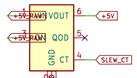
Size: A2 | Date: 2026-11-01 | **Rev: 0.1-prototype**

KiCad E.D.A. 9.0.2 | Id: 2/7

All named nets are global. Unused pins are explicitly NC.

See docs/hardware.md for design assumptions and open validation items.

F1
500mA PTC



TPS2291BDBVR



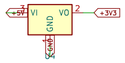
C47
1u



C48
10n



FB1
120R@100MHz



LD1117S33TR



C2
10u



C3
10u



C4
100n



C5
10u



C6
100n











ENGINEERING PROTOTYPE — NOT VALIDATED ON HARDWARE

Xenoah

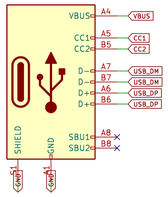
Sheet: /POWER/
File: power.kicad_sch

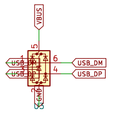
Title: VOICE PITCHER — POWER

Size: A2	Date: 2026-11-01	**Rev: 0.1-prototype**
KiCad E.D.A. 9.0.2		Id: 3/7

All named nets are global. Unused pins are explicitly NC.

See docs/hardware.md for design assumptions and open validation items.







ENGINEERING PROTOTYPE — NOT VALIDATED ON HARDWARE

Xenoah

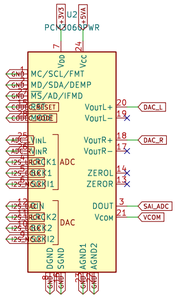
Sheet: /USB/
File: usb.kicad_sch

Title: VOICE PITCHER — USB

Size: A2	Date: 2026-11-01	**Rev: 0.1-prototype**
KiCad E.D.A. 9.0.2		Id: 4/7

All named nets are global. Unused pins are explicitly NC.

See docs/hardware.md for design assumptions and open validation items.











ENGINEERING PROTOTYPE — NOT VALIDATED ON HARDWARE

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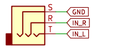
Sheet: /CODEC/
File: codec.kicad_sch

Title: VOICE PITCHER — CODEC

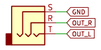
Size: A2	Date: 2026-11-01	**Rev: 0.1-prototype**
KiCad E.D.A. 9.0.2		Id: 5/7

All named nets are global. Unused pins are explicitly NC.

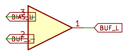
See docs/hardware.md for design assumptions and open validation items.



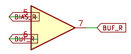
J2
3.5mm STEREO IN



J3
3.5mm STEREO OUT



U5A
TLV9062IDR



U5B
TLV9062IDR



U5C
TLV9062IDR



C23
100n

R20
10k

R21
10k



C24
470p



C25
4.7u

R22
100k

R23
100



C26
2.2n

R25
10k

R26
10k



C29
470p



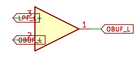
C30
4.7u

R27
100k

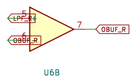
R28
100



C31
2.2n



U6A
TLV9062IDR



U6B
TLV9062IDR



U6C
TLV9062IDR



C34
100n

R30
3.3k



C35
1n

R31
100



C36
10u

R32
100k

R35
3.3k



C40
1n

R36
100



C41
10u

R37
100k

ENGINEERING PROTOTYPE — NOT VALIDATED ON HARDWARE

Xenoah

Sheet: /ANALOG/
File: analog.kicad_sch

Title: VOICE PITCHER — ANALOG

Size: A2	Date: 2026-11-01	**Rev: 0.1-prototype**
KiCad E.D.A. 9.0.2		Id: 6/7

All named nets are global. Unused pins are explicitly NC.

See docs/hardware.md for design assumptions and open validation items.



J4
OLED GND VCC SCL SDA

R40
4.7k

R41
4.7k

C42
100n



SW3
MODE

R42
10k



C43
100n

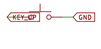


SW4
DOWN

R43
10k



C44
100n



SW5
UP

R44
10k



C45
100n



SW6
IO

R45
10k



C46
100n



D1
GREEN

R46
1k

ENGINEERING PROTOTYPE — NOT VALIDATED ON HARDWARE

Xenoah

Sheet: /UI/
File: ui.kicad_sch

Title: VOICE PITCHER — UI

Size: A2	Date: 2026-11-01	**Rev: 0.1-prototype**
KiCad E.D.A. 9.0.2		Id: 7/7